

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <string.h>
#include <time.h>

int main() {
    int pipefd[2];
    pid_t pid;

    char buffer[100];

    // Create pipe
    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    // Child = Consumer
    if (pid == 0) {

        close(pipefd[1]); // Child does not write

        printf("Child (Consumer) started.\n");

        while (read(pipefd[0], buffer, sizeof(buffer)) > 0) {
            printf("Consumer received: %s\n", buffer);
        }

        close(pipefd[0]);

        printf("Consumer finished.\n");

        exit(0);
    }

    // Parent = Producer
    else {

        close(pipefd[0]); // Parent does not read
    }
}

```

```

else {

    close(pipefd[0]); // Parent does not read

    printf("Parent (Producer) started.\n");

    const char *messages[] = {
        "Data 1",
        "Data 2",
        "Data 3",
        "Data 4",
        "Data 5"
    };

    clock_t start = clock();

    for (int i = 0; i < 5; i++) {

        write(pipefd[1], messages[i],
            strlen(messages[i]) + 1);

        printf("Producer sent: %s\n", messages[i]);
    }

    close(pipefd[1]);

    wait(NULL);

    clock_t end = clock();

    double time_taken =
        (double)(end - start) / CLOCKS_PER_SEC;

    printf("\nCommunication completed.\n");
    printf("Communication time: %f seconds\n",
        time_taken);
}

return 0;
}

```

```

snigdha@Snigdha:~$ nano practical5.c
snigdha@Snigdha:~$ gcc practical5.c -o practical5
snigdha@Snigdha:~$ ./practical5
Parent (Producer) started.
Producer sent: Data 1
Producer sent: Data 2
Producer sent: Data 3
Producer sent: Data 4
Producer sent: Data 5
Child (Consumer) started.
Consumer received: Data 1
Data 2
Data 3
Data 4
Data 5

Consumer finished.

Communication completed.
Communication time: 0.000023 seconds

```

```
snigdha@Snigdha:~$ ls -l | grep ".c"
```

```
-rwxr-xr-x 1 snigdha snigdha 16168 Aug 13 08:59 Practical.4
-rw-r--r-- 1 snigdha snigdha 1077 Aug 1 03:12 all_codes_and_outputs.txt
-rwxr-xr-x 1 snigdha snigdha 15992 Aug 5 03:12 exec
-rwxr-xr-x 1 snigdha snigdha 15992 Aug 1 03:53 exec1
-rw-r--r-- 1 snigdha snigdha 191 Jul 31 09:00 exec1.c
-rwxr-xr-x 1 snigdha snigdha 15992 Aug 5 03:14 exec2
-rw-r--r-- 1 snigdha snigdha 187 Jul 31 09:21 exec2.c
-rwxr-xr-x 1 snigdha snigdha 16048 Aug 5 03:14 exec3
-rw-r--r-- 1 snigdha snigdha 214 Jul 31 09:21 exec3.c
-rwxr-xr-x 1 snigdha snigdha 16048 Aug 1 03:53 exec4
-rw-r--r-- 1 snigdha snigdha 211 Jul 31 09:22 exec4.c
-rwxr-xr-x 1 snigdha snigdha 16120 Aug 1 03:53 exec5
-rw-r--r-- 1 snigdha snigdha 274 Aug 1 03:09 exec5.c
-rw-r--r-- 1 snigdha snigdha 0 Aug 1 03:10 exec6.c
-rwxr-xr-x 1 snigdha snigdha 16048 Aug 1 03:55 exec_family
-rw-r--r-- 1 snigdha snigdha 275 Aug 1 03:55 exec_family.c
-rw-r--r-- 1 snigdha snigdha 508 Aug 1 03:53 fork_demo.c
-rwxr-xr-x 1 snigdha snigdha 16000 Jul 31 09:00 gcc_prog1
-rwxr-xr-x 1 snigdha snigdha 16008 Jul 30 09:33 gcc_program1
-rw-r--r-- 1 snigdha snigdha 103 Jul 30 09:33 gcc_program1.c
-rwxr-xr-x 1 snigdha snigdha 16520 Jul 30 16:17 gcc_program10
-rw-r--r-- 1 snigdha snigdha 1138 Jul 30 16:17 gcc_program10.c
-rwxr-xr-x 1 snigdha snigdha 16000 Jul 30 09:34 gcc_program2
-rw-r--r-- 1 snigdha snigdha 99 Jul 30 09:34 gcc_program2.c
-rwxr-xr-x 1 snigdha snigdha 16000 Jul 30 09:34 gcc_program3
-rw-r--r-- 1 snigdha snigdha 167 Jul 30 09:34 gcc_program3.c
-rwxr-xr-x 1 snigdha snigdha 16088 Jul 30 09:35 gcc_program4
-rw-r--r-- 1 snigdha snigdha 283 Jul 30 09:35 gcc_program4.c
-rwxr-xr-x 1 snigdha snigdha 16040 Jul 30 09:35 gcc_program5
-rw-r--r-- 1 snigdha snigdha 219 Jul 30 09:35 gcc_program5.c
-rwxr-xr-x 1 snigdha snigdha 16048 Jul 30 09:36 gcc_program6
-rw-r--r-- 1 snigdha snigdha 120 Jul 30 09:36 gcc_program6.c
-rwxr-xr-x 1 snigdha snigdha 16040 Jul 30 09:36 gcc_program7
-rw-r--r-- 1 snigdha snigdha 224 Jul 30 09:36 gcc_program7.c
-rw-r--r-- 1 snigdha snigdha 734 Aug 1 03:58 mini_shell.c
-rw-r--r-- 1 snigdha snigdha 243 Aug 22 02:55 permission.c
-rw-r--r-- 1 snigdha snigdha 1573 Aug 22 03:12 practical.5
-rwxr-xr-x 1 snigdha snigdha 16264 Aug 5 04:03 practical2
-rw-r--r-- 1 snigdha snigdha 1333 Aug 5 03:57 practical2.c
-rwxr-xr-x 1 snigdha snigdha 16256 Aug 8 03:01 practical3
-rw-r--r-- 1 snigdha snigdha 949 Aug 8 02:58 practical3.c
-rwxr-xr-x 1 snigdha snigdha 16304 Aug 8 03:52 practical4
-rw-r--r-- 1 snigdha snigdha 1217 Aug 13 08:58 practical4.c
-rwxr-xr-x 1 snigdha snigdha 16472 Aug 22 03:13 practical5
-rw-r--r-- 1 snigdha snigdha 1573 Aug 22 03:13 practical5.c
-rw-r--r-- 1 snigdha snigdha 510 Jul 31 06:04 pro.c
-rw-r--r-- 1 snigdha snigdha 105 Aug 14 04:46 skill1a.c
-rw-r--r-- 1 snigdha snigdha 533 Aug 11 06:32 skill2.c
```

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {

    int pipefd[2];

    pid_t ls_pid;
    pid_t grep_pid;

    // Create pipe
    if (pipe(pipefd) == -1) {
        perror("pipe");
        return 1;
    }

    // Create first child for ls
    ls_pid = fork();

    if (ls_pid < 0) {
        perror("fork");
        return 1;
    }

    if (ls_pid == 0) {

        // Redirect stdout to pipe write end
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        // Execute ls -l
        execlp("ls", "ls", "-l", NULL);

        perror("exec ls");
        exit(1);
    }

    // Create second child for grep
    grep_pid = fork();

    if (grep_pid < 0) {
        perror("fork");
        return 1;
    }

    if (grep_pid == 0) {

        // Redirect stdin from pipe read end
        dup2(pipefd[0], STDIN_FILENO);

        close(pipefd[0]);
        close(pipefd[1]);

        // Execute grep ".c"
        execlp("grep", "grep", ".c", NULL);

        perror("exec grep");
        exit(1);
    }

    // Parent closes both pipe ends
    close(pipefd[0]);
    close(pipefd[1]);

    // Wait for both children
    waitpid(ls_pid, NULL, 0);
    waitpid(grep_pid, NULL, 0);

    printf("Pipeline execution completed.\n");

    return 0;
}

```

```

snigdha@Snigdha:~$ nano pipe_exec.c
snigdha@Snigdha:~$ gcc pipe_exec.c -o pipe_exec
snigdha@Snigdha:~$ ./pipe_exec
-rwxr-xr-x 1 snigdha snigdha 16168 Aug 13 08:59 Practical.4
-rw-r--r-- 1 snigdha snigdha 1077 Aug 1 03:12 all_codes_and_outputs.txt
-rwxr-xr-x 1 snigdha snigdha 15992 Aug 5 03:12 exec
-rwxr-xr-x 1 snigdha snigdha 15992 Aug 1 03:53 exec1
-rw-r--r-- 1 snigdha snigdha 191 Jul 31 09:00 exec1.c
-rwxr-xr-x 1 snigdha snigdha 15992 Aug 5 03:14 exec2
-rw-r--r-- 1 snigdha snigdha 187 Jul 31 09:21 exec2.c
-rwxr-xr-x 1 snigdha snigdha 16048 Aug 5 03:14 exec3
-rw-r--r-- 1 snigdha snigdha 214 Jul 31 09:21 exec3.c
-rwxr-xr-x 1 snigdha snigdha 16048 Aug 1 03:53 exec4
-rw-r--r-- 1 snigdha snigdha 211 Jul 31 09:22 exec4.c
-rwxr-xr-x 1 snigdha snigdha 16120 Aug 1 03:53 exec5
-rw-r--r-- 1 snigdha snigdha 274 Aug 1 03:09 exec5.c
-rw-r--r-- 1 snigdha snigdha 0 Aug 1 03:10 exec6.c
-rwxr-xr-x 1 snigdha snigdha 16048 Aug 1 03:55 exec_family
-rw-r--r-- 1 snigdha snigdha 275 Aug 1 03:55 exec_family.c
-rw-r--r-- 1 snigdha snigdha 508 Aug 1 03:53 fork_demo.c
-rwxr-xr-x 1 snigdha snigdha 16000 Jul 31 09:00 gcc_prog1
-rwxr-xr-x 1 snigdha snigdha 16008 Jul 30 09:33 gcc_program1
-rw-r--r-- 1 snigdha snigdha 103 Jul 30 09:33 gcc_program1.c
-rwxr-xr-x 1 snigdha snigdha 16520 Jul 30 16:17 gcc_program10
-rw-r--r-- 1 snigdha snigdha 1138 Jul 30 16:17 gcc_program10.c
-rwxr-xr-x 1 snigdha snigdha 16000 Jul 30 09:34 gcc_program2
-rw-r--r-- 1 snigdha snigdha 99 Jul 30 09:34 gcc_program2.c
-rwxr-xr-x 1 snigdha snigdha 16000 Jul 30 09:34 gcc_program3
-rw-r--r-- 1 snigdha snigdha 167 Jul 30 09:34 gcc_program3.c
-rwxr-xr-x 1 snigdha snigdha 16088 Jul 30 09:35 gcc_program4
-rw-r--r-- 1 snigdha snigdha 283 Jul 30 09:35 gcc_program4.c
-rwxr-xr-x 1 snigdha snigdha 16040 Jul 30 09:35 gcc_program5
-rw-r--r-- 1 snigdha snigdha 219 Jul 30 09:35 gcc_program5.c
-rwxr-xr-x 1 snigdha snigdha 16048 Jul 30 09:36 gcc_program6
-rw-r--r-- 1 snigdha snigdha 120 Jul 30 09:36 gcc_program6.c
-rwxr-xr-x 1 snigdha snigdha 16040 Jul 30 09:36 gcc_program7
-rw-r--r-- 1 snigdha snigdha 224 Jul 30 09:36 gcc_program7.c
-rw-r--r-- 1 snigdha snigdha 734 Aug 1 03:58 mini_shell.c
-rw-r--r-- 1 snigdha snigdha 243 Aug 22 02:55 permission.c
-rwxr-xr-x 1 snigdha snigdha 16344 Aug 22 03:15 pipe_exec
-rw-r--r-- 1 snigdha snigdha 1321 Aug 22 03:15 pipe_exec.c
-rw-r--r-- 1 snigdha snigdha 1573 Aug 22 03:12 practical.5
-rwxr-xr-x 1 snigdha snigdha 16264 Aug 5 04:03 practical2
-rw-r--r-- 1 snigdha snigdha 1333 Aug 5 03:57 practical2.c
-rwxr-xr-x 1 snigdha snigdha 16256 Aug 8 03:01 practical3
-rw-r--r-- 1 snigdha snigdha 949 Aug 8 02:58 practical3.c
-rwxr-xr-x 1 snigdha snigdha 16304 Aug 8 03:52 practical4
-rw-r--r-- 1 snigdha snigdha 1217 Aug 13 08:58 practical4.c
-rwxr-xr-x 1 snigdha snigdha 16472 Aug 22 03:13 practical5
-rw-r--r-- 1 snigdha snigdha 1573 Aug 22 03:13 practical5.c
-rw-r--r-- 1 snigdha snigdha 510 Jul 31 06:04 pro.c
-rw-r--r-- 1 snigdha snigdha 105 Aug 14 04:46 shell1.c

```